JSC370 Data Science II
Winter 2026
Midterm

GitHub: `https://github.com/ChristopherLi05/JSC370-Final-Project`

1 Introduction

This is an analysis done with publicly available data regarding the student experience at U of T. We aim to solve questions revolving around how students behave in relation to their experiences with the professor and courses. Questions include, but are not limited to:

1. How correlated is the data between INS scores and ARTSC scores (in course evals)?
2. Have professor ratings increased over time? (i.e. are they listening to feedback?)
3. Does the course evaluation scores predict response rates?
4. Does professor ratings predict dropoff %? How about course evaluations?
5. Which departments are consistently the worst?

To answer these questions, we gather our data from 3 main sources:

1. U of T Course Evaluations

   This is a survey that asks 8 to 16 statements describing the students' course learning experience. For every lecture section, responses are aggregated from the responders and published regarding the gathered statistics.

2. Rate My Prof

   This is an independent website that lets students independently and anonymously rate professors. In addition, they are also allowed to leave comments regarding their experiences.

3. U of T Timetable Data

   This is a dataset that contains timestamps and course enrollment numbers, gathered through the Timetable Builder. The dataset is used is collected from an independent third party.

2 Methods

2.1 U of T Course Evaluations

Link: Github Script
U of T course evaluations are public, though hidden from the public, by default. We use the internal API to scrape and aggregate the data. Some reverse engineering had to occur to successfully use the API.

2.2 Rate My Prof

Link: RMP Global, RMP Individual
Rate My Professor evaluations are public data, where students are able to submit several scores and an optional comment. Using the internal GraphQL API, we scrape all fields that a student may submit and store it into a CSV to be analyzed. We use VADER sentiment analysis on the comment fields in order to get a sentence sentiment score.

2.3 Course Data

U of T enrollment numbers are public data through the Timetable Builder website. In this analysis, we use the data collected by ICPRplshelp to conduct our analysis.

2.4 Data Merging

To merge these datasets together, we attempt to map each professor name to a Rate My Professor internal ID. We use a combination of exact matching and fuzzy matching to get their internal ids. As U of T has 2 school ids, professors are generally matched with up to 2 IDs.

2.5 Tools

We primarily use the `requests` library to fetch data from the apis and `pandas` to aggregate and perform operations with. We will use `matplotlib` and `seaborn` to plot, and we will use `sci-kit`, `statsmodel`, and `numpy` to do statistical analysis on the data.
We will be using a variety of models to conduct analysis on our dataset, some of which include `linear regression`, `generalized linear models`, `trees`, and other techniques.

3 Preliminary Results

Summary statistics for each calculated dataset can be found in Tables 1, 2, 3. Figures can be found following them.

Table 1: Course Evaluation Summary Statistics

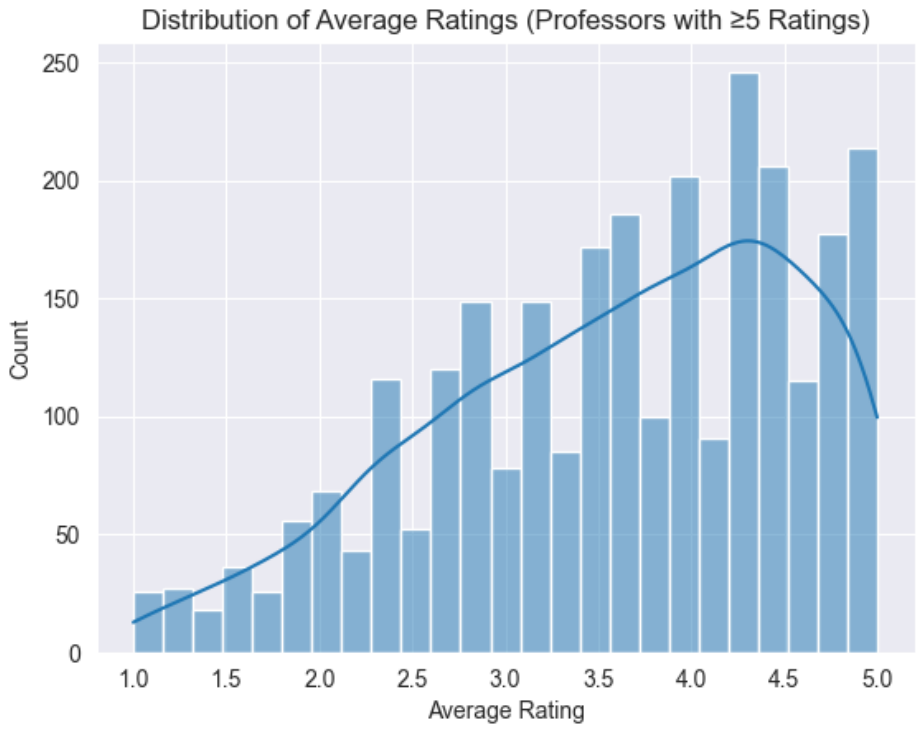
Variable	Count	Mean	Std. Dev.
year	44400	2018.645	3.778
INS1	44400	4.130	0.506
INS2	44400	4.232	0.486
INS3	44398	4.212	0.598
INS4	44400	4.021	0.529
INS5	44400	4.052	0.531
INS6	44400	3.904	0.626
ARTSC1	40860	4.282	0.566
ARTSC2	44193	3.323	0.453
ARTSC3	44200	3.914	0.644
number_invited	44387	77.001	128.956
number_responses	44400	28.794	49.993
average_response_rate	44387	0.425	0.173

Table 2: RateMyProfessors Individual Rating Statistics

Variable	Count	Mean	Std. Dev.
helpfulRating	66958	3.510	1.637
clarityRating	66958	3.517	1.632
difficultyRating	66958	3.250	1.174
wouldTakeAgain	28604	0.774	0.418
textbookUse	51350	2.120	2.297

Table 3: RateMyProfessors Professor-Level Statistics

Variable	Count	Mean	Std. Dev.
legacyId	5427	1888001	1029539
avgRating	5427	3.001	1.721
numRatings	5427	12.341	24.733
wouldTakeAgainPercent	5427	42.920	42.328
avgDifficulty	5427	2.606	1.391


Distribution of Average Ratings (Professors with ≥5 Ratings)
Count
Average Rating
0
50
100
150
200
250
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
5.0

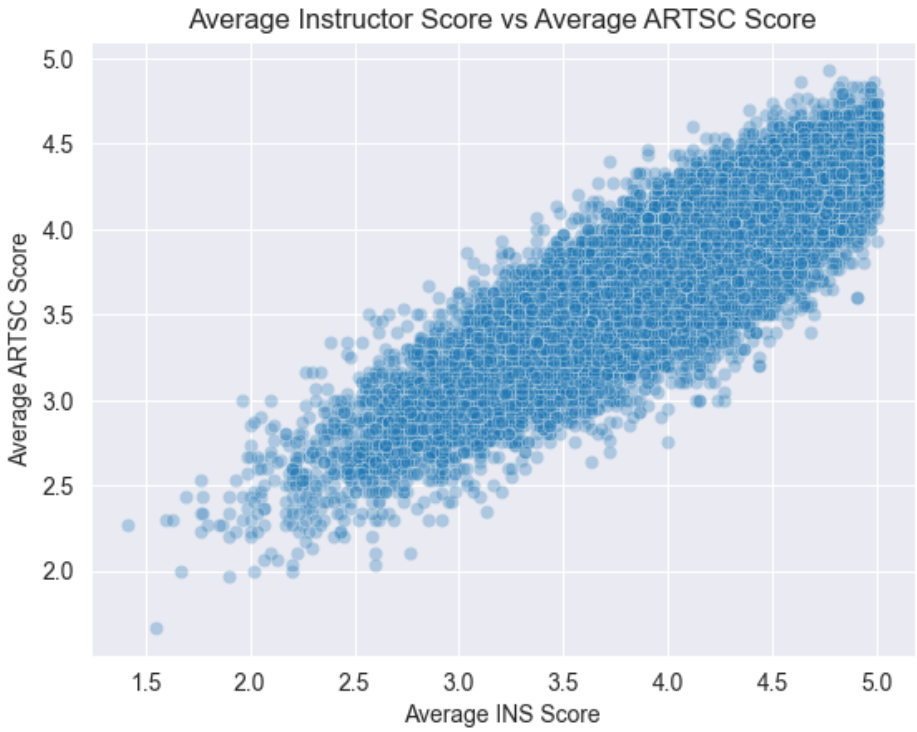

Average Instructor Score vs Average ARTSC Score
Average ARTSC Score
Average INS Score
2.0
2.5
3.0
3.5
4.0
4.5
5.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
5.0

Table 4: Binomial GLM Predicting Course Evaluation Response Rate

Variable	Coef.	Std. Err.	z	P > $\|z\|$	[0.025, 0.975]	Odds Ratio
const	-1.668	0.018	-94.06	0.000	[-1.702, -1.633]	0.189
INS1	-0.071	0.008	-9.34	0.000	[-0.085, -0.056]	0.932
INS2	-0.185	0.008	-23.09	0.000	[-0.201, -0.169]	0.831
INS3	0.053	0.007	7.43	0.000	[0.039, 0.067]	1.054
INS4	-0.020	0.009	-2.21	0.027	[-0.039, -0.002]	0.980
INS5	-0.080	0.009	-8.78	0.000	[-0.097, -0.062]	0.924
INS6	0.084	0.009	9.13	0.000	[0.066, 0.103]	1.088
ARTSC1	0.189	0.007	27.42	0.000	[0.176, 0.203]	1.208
ARTSC2	0.214	0.004	59.20	0.000	[0.207, 0.221]	1.239
ARTSC3	0.137	0.006	21.30	0.000	[0.124, 0.150]	1.147
Residual Deviance				280,087.95		
Null Deviance				292,310.63		
Pseudo R^2				0.0418		

4 Summary

4.1 Preliminary Findings

Table 1:

- The course evaluation data is large (44,400 sections)
- Instructor scores tend to be around 4.0, with low variation (around 0.5)
- There is a decent response rate (around 0.425 with 0.173 stddev), which is fairly surprising since I didn't expect so many to respond

Table 2:

- There are around 67k total reviews
- Helpful and clarity ratings average around 3.5, with high variation
- Difficulty averages 3.25, a bit surprising given how U of T is seen as more difficult than other Universities
- Would take again is fairly high, but has a large variation

Table 3:

- There are around 5,400 professors stored in here (but I believe some are stored twice)
- Average overall rating is 3.0, with a very large standard deviation (1.72)

- Most professors have few ratings (mean 12, SD 25), indicating skewed participation.

Figure 3

- Left skew, indicating professors tend to be more positively rated
- Seems to lie around 3.5 4, which matches the summary statistics

Figure 3

- Fairly significant positive correction, which makes sense - good lecture sections tend to be good

Table 4

- Many coefficients are statistically significant due to the large sample size.
- ARTSC scores have positive associations with response rate, meaning higher course level satisfaction predicts higher participation
- The pseudo R^2 is 0.0418, indicating the model explains only a small portion of the variation in response rates
- Overall, satisfaction scores do relate to response rates, but only weakly

4.2 Future Plans

The 5 questions listed are fairly straightforward to do, so in the final project we will aim to answer all 5. I plan on brainstorming more questions that I would like to ask about the data. I hope to use other ML techniques like Random Forests to answer several questions as well.